LOS/NLOS Classification and Distance Estimation for UWB Indoor Positioning Using Feature-Engineered ML and Hybrid CNN-Transformer

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***Abstract*—Accurate line-of-sight (LOS) and non-line-of-sight (NLOS) identification is essential for precise ultra-wideband (UWB) indoor positioning. We study two-path LOS/NLOS classification and path-wise distance estimation on the Decawave DWM1000 dataset. After preprocessing, a valid second path is detected in 40,864 of 41,996 usable samples (97.3%), producing an expanded two-path dataset of 83,992 rows. On the feature-engineered pipeline, XGBoost gives the best single-model AUC (0.9834), while stacked ensembling attains the highest accuracy (0.9415) with marginal AUC change. A separate raw-CIR CNN+Transformer pipeline reaches AUC 0.9821 on the original split, improving to 0.9845 with CIR augmentation, but this deep-learning result is not directly comparable to the ML results because it uses a different pipeline and split. For regression, path 1 is moderately predictable (best RMSE 1.3706 m, $R^2 = 0.6698$), whereas path 2 attains near-perfect scores because its target is derived from input features supplied to the model. These results support tree-based supervised models for this dataset, while unsupervised clustering remains weak.**

***Index Terms*—UWB, LOS/NLOS classification, channel impulse response, CNN, Transformer, indoor positioning, distance estimation**

I. Introduction

Ultra-wideband (UWB) technology has emerged as a leading solution for centimeter-level indoor positioning due to its high time resolution and resistance to multipath fading [1]. However, NLOS propagation conditions cause systematic ranging errors that severely degrade positioning accuracy. Identifying whether a signal travels via LOS or NLOS is therefore critical for mitigating these errors [2].

The channel impulse response (CIR) recorded by UWB receivers encodes rich information about the propagation environment. Traditional approaches extract hand-crafted features such as rise time, kurtosis, and energy ratios from the CIR, then apply classifiers like support vector machines (SVM), k-nearest neighbors, or random forests [3], [4]. While effective, these methods rely on domain expertise for feature selection and may miss subtle signal patterns.

Recent advances in deep learning offer an alternative: learning discriminative features directly from raw waveforms. Convolutional neural networks (CNNs) have been applied to UWB CIR classification [5], but they capture only local patterns. Transformer architectures [6] excel at modeling long-range dependencies through self-attention, and recent work has demonstrated their effectiveness for UWB NLOS detection specifically [10], [11]. This makes hybrid CNN-Transformer designs well-suited for analyzing the full CIR waveform where multipath components may be widely separated.

In this paper, we make the following contributions:

- A two-path extraction and labeling framework that extends binary LOS/NLOS classification to the two dominant signal paths in each CIR measurement.
- A feature engineering pipeline producing 25 per-path features for classical ML models, with a compact RFECV-selected subset also examined.
- A hybrid 1D-CNN + Transformer architecture that operates directly on the raw 1016-sample CIR waveform in a separate pre-computed pipeline.
- Distance estimation models for both dominant paths, interpreted with attention to the analytic constraint in the path 2 target definition.

II. Dataset and Data Preparation

A. Dataset Description

We use the publicly available UWB LOS/NLOS dataset [7] collected using Decawave DWM1000 modules across seven distinct indoor environments. The dataset contains 42,000 samples (21,000 LOS and 21,000 NLOS), each comprising 14 scalar features and a 1016-sample CIR recorded at 1 ns resolution. Table I summarizes the key features.

B. Data Preparation

Our preprocessing pipeline performs the following steps:

CIR normalization. Each CIR sample is divided by its corresponding RXPACC value to normalize the preamble accumulation effect.

Constant/metadata removal. Three raw features (CH, BITRATE, PRFR) are constant and removed. We additionally drop loader metadata (ENV_ID) to avoid using environment

TABLE I
DATASET FEATURES

Feature	Description
RANGE	Measured distance (m); also regression target
FP_IDX	First path index in CIR
FP_AMP1–3	First path amplitude estimators
STDEV_NOISE	Standard deviation of noise
CIR_PWR	Total CIR power
MAX_NOISE	Maximum noise level
RXPACC	Preamble accumulation count
FRAME_LEN	Frame length
PREAM_LEN	Preamble length
CIR0–CIR1015	1016-sample channel impulse response

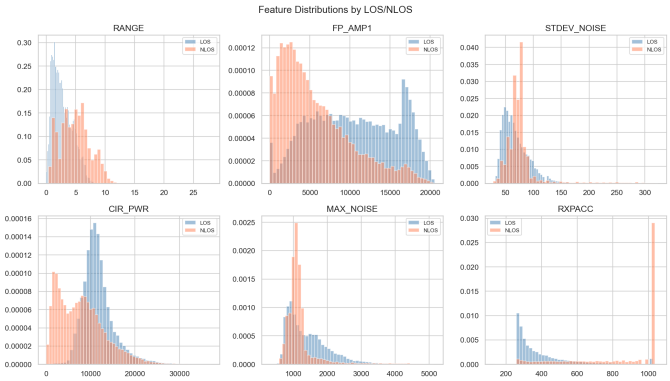

Fig. 1. Distribution of scalar features by LOS (blue) and NLOS (orange) class. Features such as CIR_PWR and FP_AMP1 show strong class separability.

identifiers as predictors, leaving the 11 scalar features used downstream.

Feature scaling. The 11 remaining scalar features are standardized using zero-mean unit-variance scaling (StandardScaler).

NaN/Inf handling. Any NaN or Inf values introduced during CIR normalization are replaced with zero.

Train/test split. After removing 4 degenerate samples, the usable set has 41,996 rows and is split stratified 80/20 into 33,596 training and 8,400 test samples.

Fig. 1 shows the distribution of key features by LOS/NLOS class, revealing clear separability for several features. Correlation analysis shows FP_AMP1–3 are highly correlated ($r > 0.9$), as are STDEV_NOISE and MAX_NOISE ($r \approx 0.95$).

III. PROPOSED METHODOLOGY

A. Two-Path Extraction

UWB CIR measurements often contain multiple propagation paths. The two shortest (dominant) paths carry the most information for positioning: if the first path is LOS, the second is necessarily NLOS (reflected); if the first path is NLOS, both paths are NLOS. Algorithm 1 describes our peak detection procedure.

After preprocessing, 41,996 samples remain usable for two-path extraction, and a second path is detected in 40,864 of them (97.3%). Fig. 2 shows example CIR waveforms with detected peaks.

Algorithm 1 Two-Path Peak Detection

Require: CIR waveform $c[0..1015]$, first path index FP_IDX
Ensure: Path indices p_1, p_2 and amplitudes a_1, a_2

$p_1 \leftarrow \arg\max_{j \in [\text{FP_IDX}-10, \text{FP_IDX}+10]} c[j]$
$a_1 \leftarrow c[p_1]$
Smooth c with a 3-sample moving average to obtain $\tilde{c}$
Find peaks in $\tilde{c}$ with height $> 0.02 \cdot a_1$, distance > 15, prominence $> 0.02 \cdot a_1$
Remove peaks within ± 15 samples of p_1
$p_2 \leftarrow$ remaining peak with maximum prominence (fallback: max amplitude)
$a_2 \leftarrow c[p_2]$ (from original unsmoothed CIR)
return p_1, a_1, p_2, a_2

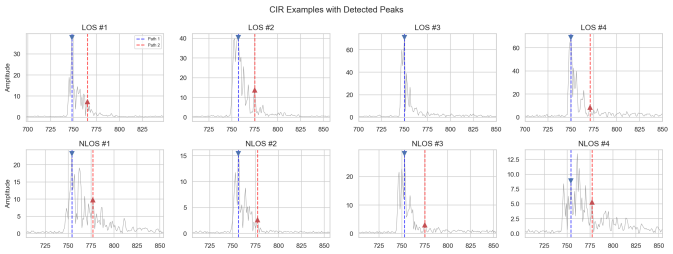

Fig. 2. Example CIR waveforms showing detected first path (triangle) and second path (circle) for LOS and NLOS conditions.

Two-path labeling. For each usable original sample, we generate two rows: path 1 retains the original LOS/NLOS label, while path 2 is always labeled NLOS: if path 1 is LOS, then path 2 is necessarily a reflected component, and if path 1 is itself NLOS, then path 2 is also NLOS. This yields an expanded dataset of 83,992 rows with a 25%/75% LOS/NLOS class imbalance.

B. Feature Engineering Pipeline

For each detected path, we compute 8 local waveform features (Table II) together with 5 full-CIR statistical descriptors, 9 shared scalar features, and 3 derived shared features, yielding a 25-dimensional feature vector per path.

Feature importance analysis indicates that amplitude- and noise-related descriptors dominate. In particular, RFECV selects a compact subset consisting of `p_path_amp` and `p_peak_to_noise`, while broader tree-based importance rankings still highlight noise-normalized peak strength as the main discriminator. SHAP analysis [16] (Fig. 4) confirms these rankings from a model-agnostic perspective.

The classical ML study includes Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF) [9], Gradient Boosted Trees (GBT), and XGBoost (XGB) [8], followed by simple averaging and stacked ensembles over the strongest tree-based models.

C. End-to-End Deep Learning Pipeline

We propose a hybrid 1D-CNN + Transformer architecture that operates directly on the raw 1016-sample CIR waveform. Algorithm 2 describes the training procedure.

TABLE II
KEY LOCAL PER-PATH ENGINEERED FEATURES (8 OF THE 13 PER-PATH DESCRIPTORS)

Feature	Description
path_idx	Peak sample index in CIR
path_amp	Peak amplitude
rise_time	Samples from 10% threshold to peak
decay_time	Samples from peak to 10% threshold
kurtosis_local	Kurtosis of ±15-sample window
energy_ratio	Local energy / total CIR energy
peak_to_noise	Peak amplitude / STDEV_NOISE
amplitude_ratio	This path amp / other path amp

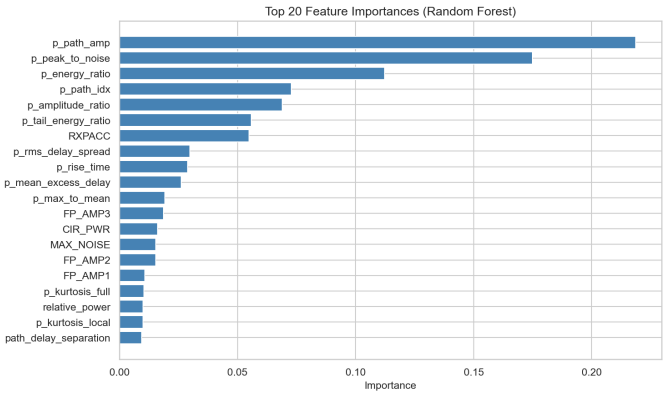

Fig. 3. Top 15 features ranked by Random Forest Gini importance. Signal power and noise-related features dominate.

Architecture. The model consists of three components:

1) CNN encoder: A multi-scale first convolutional stage (parallel kernels 3/7/15) is followed by an initial MaxPool(2) that halves the sequence from 1016 to 508, then two residual Conv1D blocks [14] (kernels 5 and 3) with max-pooling progressively reduce this to 127 feature vectors of dimension 128.

2) Transformer encoder: Two Transformer encoder layers with 4-head self-attention model global dependencies across the CNN-encoded sequence. Sinusoidal positional encoding preserves temporal ordering. The self-attention mechanism allows the model to relate distant CIR components, capturing multipath relationships that span the full waveform.

3) Classification head: Global average pooling reduces the transformer output to a single 128-dimensional vector, which is concatenated with 11 scaled scalar features and passed through a two-layer MLP (hidden size 64) to produce a binary logit.

The DL model is trained on the original balanced 42,000-sample dataset (not the two-path expanded set), using an 80/20 stratified split in a separate pre-computed raw-CIR pipeline.

IV. EXPERIMENTAL RESULTS

A. Classification Results

Table IV summarizes the supervised classification results. The feature-engineered ML models and ensembles are evaluated on the two-path expanded test set, while the CNN+Transformer results come from a separate raw-CIR pipeline on the original split. The latter should therefore be read as a related reference point rather than a direct one-to-one comparison.

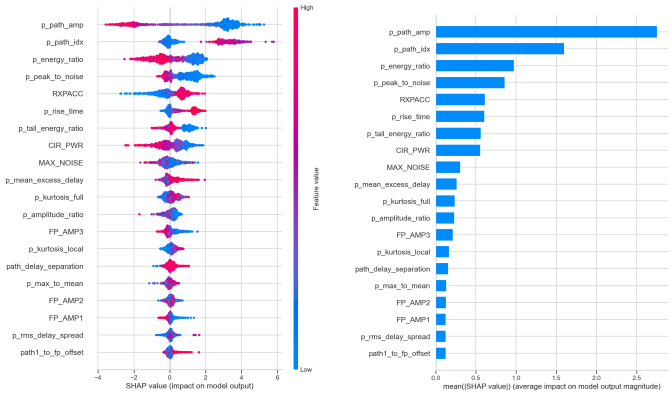

Fig. 4. SHAP feature impact: beeswarm plot (left) showing per-sample feature contributions and mean absolute SHAP values (right) confirming Gini-based rankings.

TABLE III
CNN-TRANSFORMER HYPERPARAMETERS

Parameter	Value
CNN channels	$32 \rightarrow 64 \rightarrow 128$
CNN kernels	3/7/15 (parallel), then 5 and 3
Transformer layers	2
Attention heads	4
Feedforward dim	256
MLP hidden	64
Dropout	0.1
Optimizer	Adam [15] (lr = 1×10^{-3})
Loss	BCEWithLogitsLoss
Label smoothing	$\varepsilon = 0.05$ ($y' = 0.975/0.025$) [12]
Batch size	256
Early stopping	patience = 5
Device	Auto-select (CUDA/MPS/CPU)

Among directly comparable feature-engineered models, XGBoost is best by AUC (0.9834), while stacked ensembling gives the highest accuracy (0.9415) with essentially no AUC gain over XGBoost alone. SVM performs weakest, likely because the subsampled training set and current feature representation are ill-suited for a margin-based classifier. In the separate raw-CIR pipeline, CIR augmentation improves the CNN+Transformer more noticeably than SMOTE [13] improves the ML models; however, this DL-vs-ML comparison must remain qualified because the underlying splits and inputs differ. Fig. 5 and Fig. 6 show representative ROC and confusion-matrix views for the main baseline models.

B. Distance Estimation Results

Table V presents regression performance for both signal paths. Path 1 uses the original test split, while path 2 uses the 8,193 valid path 2 test samples retained after preprocessing and two-path extraction.

Path 1 regression is materially harder than classification because the target RANGE includes environment-dependent NLOS bias [3], [4], with the best results coming from

Algorithm 2 CNN-Transformer Training

Require: Training set $\{(\mathbf{c}_i, \mathbf{s}_i, y_i)\}_{i=1}^{N}$ (CIR, scalars, label)
Ensure: Trained model f_θ

```
Initialize f_θ with random weights
best_loss ← ∞; patience_count ← 0
for epoch = 1 to 50 do
    for each mini-batch B do
        y'_B ← y_B · 0.95 + 0.025 {label smoothing, ε=0.05}
        ŷ ← f_θ(c_B, s_B)
        L ← BCEWithLogits(ŷ, y'_B)
        Update θ via Adam with ∇_θ L
    end for
    Compute validation loss L_val
    if L_val < best_loss then
        best_loss ← L_val; save θ; reset patience
    else
        patience_count ← patience_count + 1
    end if
    if patience_count ≥ 5 then
        break (early stopping)
    end if
end for
return f_θ with best validation weights
```

TABLE IV
CLASSIFICATION PERFORMANCE

Model	Accuracy	AUC	Test Set
Logistic Regression	0.9199	0.9694	Expanded
SVM	0.6523	0.8734	Expanded
Random Forest	0.9386	0.9815	Expanded
Gradient Boosted Trees	0.9311	0.9819	Expanded
XGBoost	0.9404	0.9834	Expanded
Ensemble Avg	0.9402	0.9831	Expanded
Ensemble Stacked	0.9415	0.9830	Expanded
CNN+Transformer	0.9360	0.9821	Original split
CNN+Transformer + CIR Aug	0.9421	0.9845	Original split

RF/XGB-style tree ensembles at roughly 1.37 m RMSE and $R^2 \approx 0.67$. Path 2 appears much easier, but this must be interpreted carefully: the target is defined from $\texttt{RANGE} + (\texttt{path2_idx} - \texttt{FP_IDX}) \cdot c$, while RANGE is also supplied as an input. The near-perfect path 2 scores therefore reflect a strong analytical constraint in the target construction rather than an entirely independent prediction task. Fig. 7 shows predicted vs. actual distance scatter plots.

V. DISCUSSION

A. ML vs. Deep Learning Comparison

The principal supervised finding is that feature-engineered tree models perform well on the expanded two-path task: XGBoost reaches AUC 0.9834, and stacked ensembling yields only marginal improvement. The CNN+Transformer also performs strongly, reaching AUC 0.9821 on the original split and 0.9845 with CIR augmentation, but this ML-vs-DL comparison is not direct because the deep model is trained and evaluated in a separate raw-CIR pipeline on the original split. Fig. 9 summarizes the model comparison.

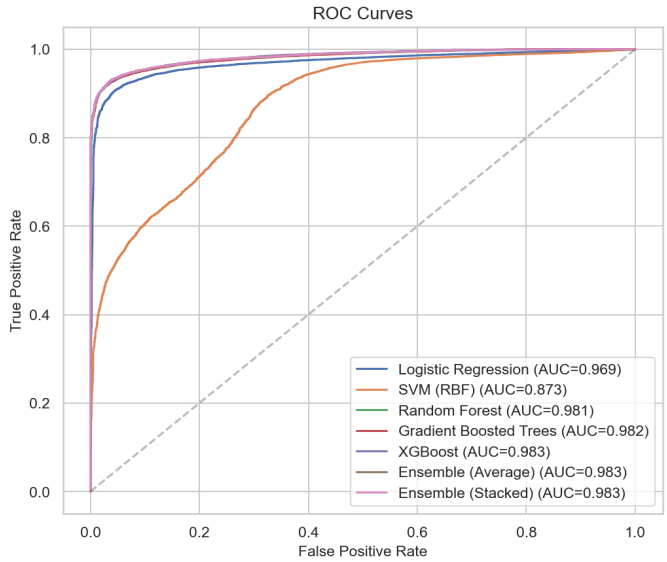

Fig. 5. ROC curves for selected baseline classification models. Tree-based ML and CNN+Transformer achieve similarly high AUC values.

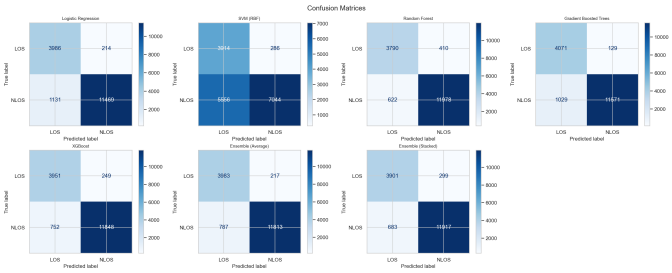

Fig. 6. Confusion matrices for selected baseline classification models.

Accordingly, we treat the DL results as complementary evidence that raw-CIR learning is viable, rather than as evidence that DL surpasses ML on an identical benchmark. Within the notebook experiments, SMOTE has negligible effect on the ML pipeline, whereas CIR augmentation produces a somewhat more visible gain for the separate CNN+Transformer pipeline.

B. Interpretability via Attention

A key advantage of the CNN-Transformer is interpretability through attention weight visualization. Fig. 10 shows a representative attention pattern for sample CIR waveforms. Qualitatively, elevated attention often appears near first-arrival and multipath regions, offering suggestive evidence about which CIR segments influence predictions. This complements the feature-importance analysis of the classical models.

C. Error Analysis

Classification errors primarily occur in ambiguous cases where NLOS signals have strong direct-path-like characteristics or where LOS signals experience significant multipath interference. The gap between LR and the tree-based models

TABLE V
DISTANCE ESTIMATION PERFORMANCE

Path	Model	RMSE (m)	R^2
Path 1	Ridge	1.6164	0.5407
	Random Forest	1.3706	0.6698
	GBT	1.3872	0.6617
	XGBoost	1.3726	0.6688
Path 2	Ridge*	0.0000	1.0000
	Random Forest	0.4696	0.9997
	GBT	0.2889	0.9999
	XGBoost	1.2176	0.9981

*Perfect score: target is a near-deterministic linear function of inputs.

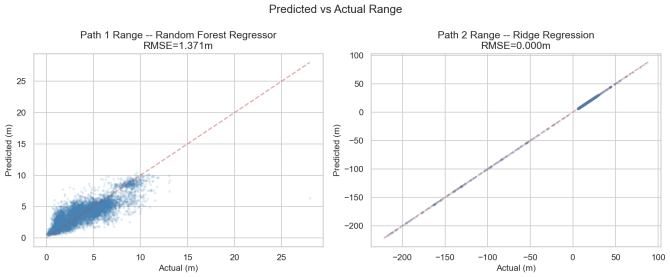

Fig. 7. Predicted vs. actual distance for path 1 (left) and path 2 (right) regression models.

suggests that the decision boundary is nonlinear, while the weak SVM result suggests that this particular feature representation and tuning are not robust enough for a margin-based classifier.

For distance estimation, path 1 remains the more meaningful regression task because the measured range includes environment-dependent bias that is not trivially recoverable from a single formula. By contrast, path 2 regression benefits from the direct geometric relationship embedded in the target definition and should not be overstated.

Unsupervised clustering is weak. KMeans reaches only 0.5906 test accuracy with ARI 0.0269, and DBSCAN fails to discover any clusters, labeling 100% of samples as noise. These results suggest that LOS/NLOS separation in this feature space is not readily achievable through unsupervised methods alone.

From a practical standpoint, the CNN-Transformer operates on the full 1016-sample CIR plus 11 scalar features, avoiding manual feature engineering, whereas the tabular models require the 25-dimensional engineered feature vector.

VI. CONCLUSION

We presented a concise UWB LOS/NLOS classification and distance-estimation study using a feature-engineered two-path pipeline together with a separate raw-CIR CNN+Transformer reference pipeline. The two-path extraction stage finds a valid second path in 97.3% of usable preprocessed samples and expands the supervised dataset to 83,992 rows.

The key findings are as follows: (1) among directly comparable ML models, XGBoost gives the best single-model

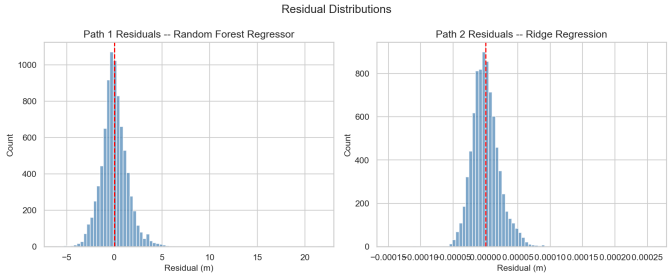

Fig. 8. Regression residual distributions. Path 1 shows a slight positive bias from NLOS ranging error; path 2 residuals are tightly clustered around zero, consistent with the analytically constrained target.

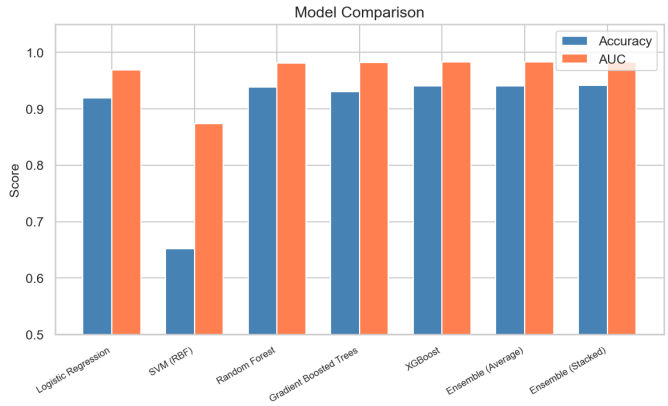

Fig. 9. Comparison of classification metrics across all models.

AUC (0.9834), while stacked ensembling gives the highest accuracy (0.9415) with marginal AUC change; (2) the CNN+Transformer is competitive, and CIR augmentation improves it further, but this DL result comes from a separate original-split raw-CIR pipeline and is not a strict one-to-one comparison with the ML results; (3) path 1 regression is moderately accurate, whereas path 2 regression is analytically constrained by the way its target is defined; and (4) clustering results are weak and should not be used to claim strong unsupervised separability.

A. Future Work

Several directions merit further investigation: (1) Ensemble methods combining ML and DL predictions; (2) Cross-environment generalization studies using leave-one-environment-out validation; (3) Model compression and quantization for real-time deployment on embedded UWB receivers; (4) Extension to multi-class scenarios distinguishing between different NLOS propagation mechanisms (diffraction, reflection, scattering).

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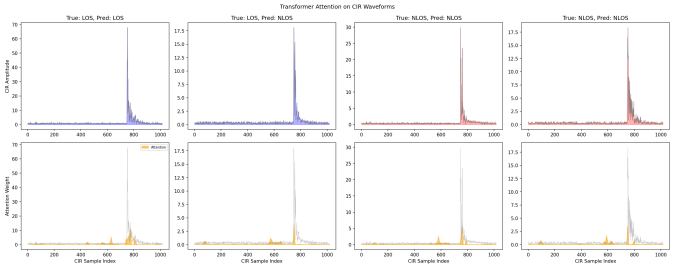

Fig. 10. Transformer attention weights overlaid on a CIR waveform. High attention concentrates near the first path and multipath components.

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